# Predicting the Risk of Obesity in the United States

**Caleb Kim** [*1] **Daniel Song** [*1] **Christina Yang** [*1]

## Abstract

This study examines how individual health behaviors, demographic characteristics, and food environmental conditions can be leveraged for the purpose of predicting adult obesity in the United States. Through utilizing a combined dataset constructed from the Behavioral Risk Factor Surveillance System (BRFSS), the USDA Food Environment Atlas, and the U.S. Census Bureau's American Community Survey, we developed a predictive modeling approach to classify obesity status, which is defined as having a body mass index (BMI) of 30 or higher. The dataset required extensive preprocessing, including removing implausible BMI values, handling missing data, and standardizing all predictors. We evaluated three classification models including k-Nearest Neighbor, Decision Tree, and Logistic Regression. Each model went through hyperparameter tuning through stratified five-fold cross-validation prior to final testing. The cleaned dataset which consisted of 148,558 individuals was split into an 80-20 train-test partition. This preserved proportional obesity prevalence across subsets. Across models, predictive performance was modest but consistent. Test accuracies ranged from 68.45% to 69.44% and ROC AUC scores were between 0.613 and 0.621. Overall, the Decision Tree classifier achieved the strongest results, outperforming kNN and Logistic Regression in both accuracy and AUC. This finding suggests that nonlinear interactions among behavioral and environmental factors carry meaningful predictive signals. Nevertheless, all models demonstrated limitations in identifying obese individuals, which was reflected by low recall and F1-scores for the positive class. This pattern can be attributed to class imbalance and considerable overlap in predictor distributions. Through feature importance analysis, it was revealed that physical activity, food-access indicators, racial composition, grocery and restaurant density, income, and smoking behaviors were the strongest contributors to obesity prediction, emphasizing the combined influence of personal health behaviors and broader structural determinants. In conclusion, the findings indicate that while machine learning models can detect broad patterns associated with obesity risk, the complexity of behavioral and environmental influences limits predictive precision at the individual level. These results highlight the need for refined modeling approaches and more specific data to improve sensitivity. They also reaffirm that obesity is shaped together by lifestyle choices, socioeconomic conditions, and community food environments.

## 1. Data Description

The purpose of this project is to predict obesity risk through looking at factors including lifestyle, demographics, and the environment. Obesity, defined as a Body Mass Index (BMI) of 30 or higher, is a major public health challenge in the United States. Its prevalence is due to both individual behaviors and structural conditions in the surrounding environment. So, the main research question we aim to answer is: To what extent can adult obesity status be predicted using individual-level behavioral and demographic factors combined with state-level food environment indicators, and which predictors contribute most to model performance? Our dataset is constructed from multiple raw data sources that require substantial preparation and integration.

The primary source of individual health information is the Behavioral Risk Factor Surveillance System (BRFSS), which is an annual survey conducted by the Centers for Disease Control and Prevention (CDC). This survey collects nationwide self-reported health behaviors, including diet, physical activity, smoking, alcohol use, and basic demographic characteristics such as age, sex, race or ethnicity, and income. In BRFSS 2017, BMI is provided as the de-

[*]Equal contribution [1]Department of Data Science, University of Virginia, Charlottesville Virginia, United States. Correspondence to: Caleb Kim <qch2ev@virginia.edu>, Daniel Song <yen2pn.2@virginia.edu>, Christina Yang <gca9aa@virginia.edu>.

*Proceedings of the 42nd International Conference on Machine Learning*, Vancouver, Canada. PMLR 267, 2025. 

rived variable _BMI5, which stores BMI multiplied by 100. We converted this value to standard BMI units by dividing by 100 and removed biologically implausible observations, by restricting BMI to the range 15 to 60. Obesity was then defined as a binary outcome equal to 1 when BMI was at least 30 and 0 otherwise. BRFSS provides a large, rich foundation of individual-level records that link personal health behaviors with obesity outcomes.

In order to capture environmental context, we merged state-level indicators from the USDA Food Environment Atlas. We included measures of the retail food landscape including grocery store counts and per-capita availability (GROC14, GROCPTH14), supercenters and convenience stores (SUPERC14, CONVS14, and their per-capita measures), and SNAP-authorized retailers (SNAPS16, SNAPSPTH16). We also incorporated restaurant environment measures including fast food and full-service restaurant counts and density (FFR14, FFRPTH14, FSR14, FSRPTH14) and per-capita sales proxies (PC_FFRSALES12, PC_FSRSALES12). Food access was then represented using low-access percentages from 2015 (PCT_LACCESS_POP15 and subgroup variants). Because these indicators are environmental and not individual-level, we aligned them to BRFSS by aggregating the USDA data to the state level, summing total outlet counts and averaging per-capita and percentage measures, then merging by state abbreviation.

We merged U.S. Census Bureau's American Community Survey to include the state names. The main focus remained on merging BRFSS health behaviors with USDA state-level food environment characteristics. This streamlined approach ensured that all datasets aligned cleanly at the state level, while still capturing both individual and environmental determinants of obesity.

The outcome variable of interest is obesity status, which is defined as a binary classification based on BMI. The final feature set consists of individual-level BRFSS predictors (age, sex, race or ethnicity indicators, income, exercise, physical activity minutes, smoking history and current smoking, alcohol frequency and binge drinking, and fruit and vegetable intake) and state-level USDA predictors (food outlet counts and per-capita density measures, food access percentages, and restaurant environment measures). The target variable is obesity status derived from BMI.

Preparing these data required explicit cleaning and recoding steps. In BRFSS, several variables use numeric codes to represent nonresponse. We converted "don't know" and "refused" responses to missing values using the codebook conventions for each field. For example, INCOME2 codes 77 and 99, and several behavioral items use 7 and 9 for don't know and refused. We also converted derived BMI from _BMI5 into standard BMI units and excluded implausible BMI values by restricting BMI to 15 through 60. We then transformed multiple frequency-coded variables into interpretable continuous measures. Specifically, ALCDAY5 was converted into an estimated number of drinking days per month, and FRUIT2 and VEGETAB2 were converted into approximate daily intake frequencies. Several predictors were recoded into analysis-ready formats including binary indicators for sex, exercise, and smoking status, and indicator variables for major race or ethnicity categories. After merging the state-level USDA indicators with individual BRFSS records, we addressed remaining missingness using variable-specific imputations: continuous behavioral variables were imputed using the mean, binary predictors using the mode, and ordinal income using the median. Continuous predictors were later standardized for model training.

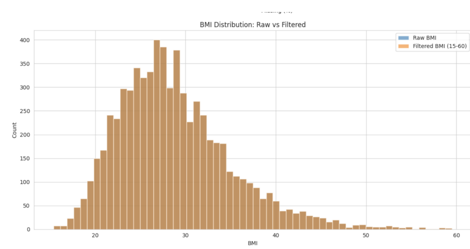


*Figure 1.* BMI Distribution

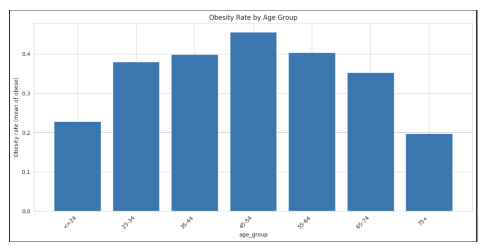


*Figure 2.* Obesity Rate by Age Group

The volume and complexity of the BRFSS data also increased the need for efficient filtering and preprocessing. With more than 148,000 observations retained after cleaning, careful handling of missingness and feature engineering was necessary to ensure a reliable analytical file. Through constructing BMI values, excluding unrealistic cases, recoding categorical responses, and merging environmental datasets at the state level, we created a structured dataset suitable for predictive modeling. Since data from BRFSS can include extreme or unrealistic responses, we applied clear rules to remove outliers. Missing values were handled through imputation, and continuous features were standardized prior to model fitting.

Through this process, we created a unified dataset that links individual health behaviors with the structural characteristics of the surrounding food environment. This allows us to examine not only how personal factors influence obesity but also how broader community contexts shape health outcomes. Through reading, cleaning, and merging these raw sources, we prepared a reliable foundation for the predictive modeling and analysis that follow.

## 2. Pre-Analysis Plan

### 2.1. Method Overview

Our main objective is to create predictive models that integrate socioeconomic, environmental, and individual data to estimate the likelihood of adult obesity in the United States. We will build a supervised learning pipeline that consists of feature engineering, data preprocessing, model training, and model evaluation in order to accomplish this. Predictors include individual-level health behaviors and demographic traits, as well as environmental conditions at the community level. The outcome variable, obesity status, is defined as a binary indicator based on Body Mass Index (BMI $\geq$ 30).

The project will go through a number of stages. The first phase is data wrangling. This involves cleaning and organizing raw survey and contextual data, dealing with inconsistent or missing entries, and transforming categorical variables into numerical formats appropriate for modeling. Because the environmental datasets in the final implementation were aggregated to the state level rather than the county level, merging across datasets relies on consistent state identifiers rather than FIPS codes. Exploratory data analysis (EDA) will then be used to identify outliers, visualize variable distributions, summarize important aspects of the data, and investigate possible connections between predictors and obesity outcomes. Following the complete preparation of the data, the modeling phase will concentrate on creating classification models that forecast the risk of obesity using the processed dataset. To find strategies that offer the best balance between interpretability and predictive accuracy, both baseline and more adaptable models will be tested.

Once the models are trained, they will be validated and evaluated to measure performance and determine generalizability to new data. Accuracy-based metrics will be used to compare various approaches, along with ROC curves and AUC scores to assess classification performance. The final step in the model interpretation process will be to summarize the environmental and individual factors that have the greatest impact on the risk of obesity. This methodical approach preserves a transparent and repeatable modeling process while relating individual health data to more general social and environmental factors.

### 2.2. Models to Be Used and Justification

In order to predict obesity status, we will implement models including k-Nearest Neighbor (kNN), Decision Tree classification, and Logistic Regression. These approaches complement one another when it comes to interpretability and flexibility, making them suitable for modeling a complex outcome like obesity status. First, the kNN model will serve as an intuitive baseline for classification, as it does not assume a fixed functional form between predictors and outcomes. Instead it works by assigning a label to each observation based on the majority class among its closest neighbors in the space, allowing the model to capture local patterns and nonlinear relationships across behavioral and demographic factors.

Decision trees divide the predictor space into a hierarchy of decision rules that identify important variable interactions. The visual structure of decision trees allows for clear interpretation of how different conditions, such as limited food access or low physical activity, combine to increase the likelihood of obesity.

Logistic Regression provides a more classical parametric model that directly estimates the relationship between predictors and the log-odds of obesity. Because logistic regression offers interpretable coefficients and well-understood statistical properties, it serves as an essential comparison point alongside the nonparametric and tree-based models.

Together, these three models offer a balanced combination of interpretability and predictive power and are suitable for explaining findings to a public-health audience.

### 2.3. Model Training Procedure

To train the models, we will prepare the data for analysis, select the appropriate model hyperparameters, and then tune the models to optimize performance. Prior to training, all predictors including demographic variables, behavioral measures, and food environment indicators will be standardized to have a mean of zero and a standard deviation of one. This ensures that all variables contribute equally to distance-based algorithms such as kNN. Because the final dataset contains no nominal categorical variables after preprocessing, one-hot encoding is not required; instead, the dataset consists entirely of numeric features. Missing values in continuous variables will be replaced using mean imputation, while binary or categorical indicators will be imputed using their most frequent category. Implausible BMI values and extreme survey responses will be removed to ensure that the model is trained on realistic and representative cases.

Once the data is cleaned, the dataset will be randomly divided into training and testing subsets, with 80% allocated for training and 20% for testing. Stratified sampling will be used to ensure balanced representation of obese and non-obese individuals. Within the training set, 5-fold stratified cross-validation will be used to identify optimal hyperparameters for each model. For kNN, the number of neighbors and weighting schemes will be evaluated. For the Decision

Tree classifier, parameters such as maximum depth, splitting criteria, and minimum leaf sizes will be optimized through grid search. For Logistic Regression, hyperparameters such as regularization strength (C), solver choice, and maximum iterations will be tuned to achieve stable convergence and maximize classification accuracy.

All training and analysis will be conducted in Python using libraries such as pandas and numpy for data management and scikit-learn for modeling and hyperparameter optimization. The configurations, random seeds, cross-validation results, and final model parameters will be documented to ensure reproducibility.

### 2.4. Validation Plan

We will use a validation framework that divides the dataset into training and testing subsets by random allocation in order to evaluate model generalization. While the training phase fits model parameters, the testing phase provides an impartial evaluation of predictive performance on unseen observations. This partitioning approach ensures that we can distinguish true signal from overfitting.

Classification accuracy will serve as a core evaluation metric, supplemented by ROC curves and AUC to assess the models' ability to distinguish between obese and non-obese individuals. Confusion matrices will be used to visualize classification errors across categories and identify systematic patterns of misclassification. We will also compare performance across different model configurations, such as varying values of k in kNN or different regularization levels in logistic regression, to diagnose potential overfitting or underfitting.

After establishing adequate performance on the training data, the models will be evaluated on the held-out test set to estimate real-world generalization accuracy. Consistency between training and testing metrics indicates successful generalization, whereas substantial discrepancies may require adjustments to preprocessing or hyperparameter selection. Throughout the analysis, data handling, model fitting, and validation procedures will be conducted transparently to support replicability and ensure adherence to accepted modeling practices.

## 3. Results

### 3.1. Model Implementation and Training

The cleaned dataset contained 148,558 individuals after preprocessing, and the modeling process followed the pipeline described in the pre-analysis plan. The data was divided into an 80–20 stratified training and testing split in order to preserve the proportion of obese and non-obese individuals. This resulted in 118,846 training observations and 29,712 testing observations, with both subsets maintaining an identical obesity prevalence of 30.55%. Before fitting any models, all predictor variables were standardized using z-score normalization, so that features with larger numerical ranges such as age or income would not dominate the distance calculations for methods like kNN. For example, the AGE variable had a training-set mean of 55.13 and a standard deviation of 17.42 before standardization, which transformed to approximately zero mean and unit variance afterward.

Three supervised learning models were trained: k-Nearest Neighbors, a Decision Tree classifier, and Logistic Regression. Each model underwent hyperparameter tuning through five-fold stratified cross-validation. For kNN, the search grids included multiple values of k, weighting schemes, and a fixed Euclidean distance metric. For the Decision Tree, they included a wide range of tree depths, leaf sizes, and splitting criteria. And for Logistic Regression, they included various regularization strengths, solvers, and iteration limits. Cross-validation ensured that each selected model represented its most stable and accurate configuration on the training data before final evaluation.

### 3.2. Evaluation Benchmarks

The three optimally tuned models were evaluated on the held-out test set using accuracy, ROC AUC, confusion matrices, and classification reports. These metrics allowed for a comprehensive assessment of how well each classifier generalized to unseen data. Accuracy reflected the overall proportion of correct predictions, while ROC AUC provided a measure of the model's ability to rank obese against non-obese individuals. The confusion matrices revealed how well each classifier distinguished between the two classes, particularly given the imbalance toward non-obese cases.

The best-performing configurations were a kNN model with fifteen neighbors, uniform weighting, and Euclidean distance, a Decision Tree with a maximum depth of seven and a minimum leaf size of two, and a Logistic Regression model using an L2 penalty with a regularization parameter of 0.01 and a maximum of 300 iterations. Cross-validation accuracies ranged from approximately 0.682 to 0.695, which indicated consistent performance across models and folds.

### 3.3. Overall Model Performance

Across all three classifiers, performance on the test set was relatively similar, with accuracies clustering in the 0.68 to 0.70 range. The Decision Tree achieved the highest accuracy at 69.44 percent and the highest ROC AUC at 0.621. Logistic Regression followed with a test accuracy of 69.42 percent and an AUC of 0.613, while kNN achieved 68.45 percent accuracy and an AUC of 0.614. These results indicate that the Decision Tree captured slightly more predictive signal than the other models, although the differences were modest.

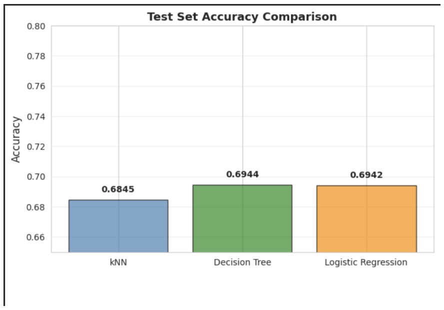


*Figure 3.* Test Set Accuracy Comparison

The accuracy and ROC AUC bar charts visually reinforced this observation, as the Decision Tree bars were consistently slightly higher than those of kNN and Logistic Regression. The ROC curves further illustrated this pattern, as the Decision Tree curve remained above the others across most thresholds, demonstrating its superior ranking capability. Although none of the models achieved exceptionally high discriminative performance, each performed noticeably better than random guessing, and their curves showed consistent separation between obese and non-obese individuals.

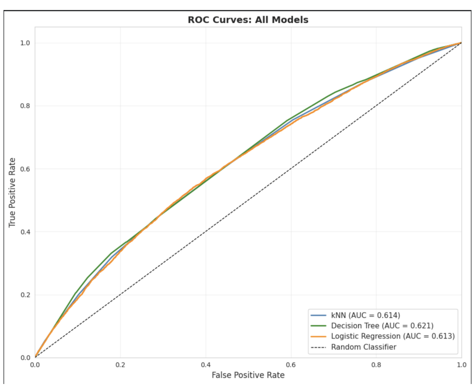


*Figure 4.* ROC Curves: All Models

### 3.4. Model-Specific Performance Diagnostics

The confusion matrices highlighted a critical aspect of model behavior. All three models performed strongly on the non-obese class, with recall values exceeding 90 percent, but they struggled to correctly identify obese individuals. Logistic Regression performed the poorest in this respect, correctly identifying fewer than 3 percent of obese individuals, while kNN achieved approximately 20 percent recall for the obese class, and the Decision Tree achieved slightly above eleven percent. The classification reports confirmed these findings, as the positive class (obese) exhibited notably low F1-scores across all models.

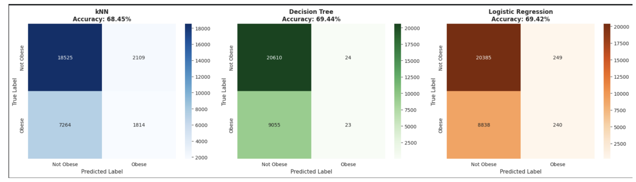


*Figure 5.* Confusion Matrices

These patterns suggest that class imbalance, combined with substantial overlap in the predictor distributions between obese and non-obese individuals, significantly affected model sensitivity. Logistic Regression, in particular, defaulted heavily toward the majority class despite tuning the regularization parameter. The Decision Tree provided somewhat better sensitivity, indicating that nonlinear splits captured additional distinctions that linear decision boundaries could not.

### 3.5. Feature Importance Analysis

An examination of feature importance helped clarify the variables most strongly associated with obesity likelihood. The Decision Tree importance scores and the Logistic Regression standardized coefficients jointly revealed that both behavioral and environmental characteristics contributed meaningfully to prediction. Exercise frequency emerged as the most influential predictor for both models, with reduced physical activity associated with higher obesity risk. Features capturing limited food access, such as the percentage of the population living in low-access areas, also showed strong contributions and were particularly elevated in the Decision Tree analysis.

Race-specific indicators, especially the proportion of individuals identifying as Black, appeared as notable predictors, consistent with documented disparities in obesity prevalence. Measures of grocery store availability, full-service restaurant density, income levels, and physical activity minutes additionally ranked near the top of both importance lists. Logistic Regression coefficients clarified the directionality of associations, whereas the Decision Tree revealed nonlinear interactions among environmental and behavioral factors. Together, the two models suggested that obesity risk is shaped jointly by lifestyle, socioeconomic status, and broader structural food-environment conditions.

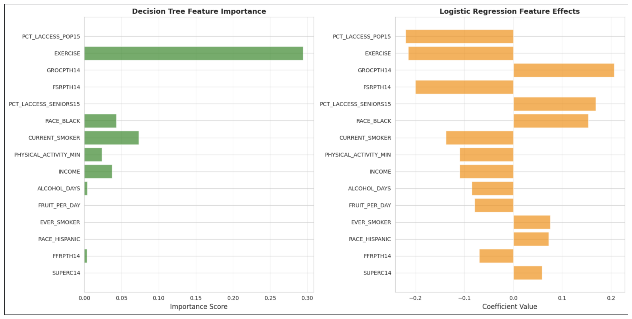


*Figure 6.* Feature Importance Comparison

### 3.6. Interpretation and Takeaways

The early findings demonstrate that obesity risk can be predicted with modest accuracy using a combination of demographic, behavioral, and environmental features. While overall accuracy was reasonably high, all models struggled with sensitivity to the obese class, suggesting that the distributions of predictors for obese and non-obese individuals substantially overlap. The Decision Tree consistently produced the strongest results and therefore appears to capture complex interactions that the other models could not fully detect. The analysis further confirms that environmental indicators such as food access and store density meaningfully contribute to prediction, reinforcing the importance of structural determinants of health. Behavioral factors remained the strongest predictors across all models, which aligns with public-health literature emphasizing the roles of physical activity and diet-related behaviors.

## Conclusion

The objective of this project was to integrate behavioral, demographic, socioeconomic, and environmental data from several large-scale public datasets to create an interpretable, data-driven framework for predicting adult obesity in the United States. We created a single analytical file that represents both individual health behaviors and the more general structural factors known to influence obesity outcomes by combining state-level indicators from the USDA Food Environment Atlas and the American Community Survey (ACS) with individual-level responses from the Behavioral Risk Factor Surveillance System (BRFSS). Extensive cleaning was necessary to prepare these heterogeneous sources, which included standardizing all continuous predictors, eliminating implausible BMI values, recoding categorical survey responses, changing frequency-based behavioral variables, and imputing missingness using variable-appropriate techniques. The final feature matrix was consistent, comprehensible, and appropriate for repeatable machine-learning analysis thanks to this stringent preprocessing pipeline.

Although the performance ceiling is intrinsically constrained by the outcome's complexity, modeling results showed that obesity risk is partially predictable from the available data. Test accuracies for k-Nearest Neighbors, Logistic Regression, and Decision Tree classifiers were consistently between 68% and 70%, and ROC AUC values varied between 0.613 and 0.621. The Decision Tree performed the best overall, indicating that nonlinear relationships between environmental and behavioral factors significantly influence prediction. However, all models had trouble being sensitive to the obese class, as evidenced by consistently low recall and F1-scores. The confusion matrices showed that while classifiers did well on the non-obese class, they often misclassified obese people. Significant overlap in predictor distributions between the two classes is highlighted by this pattern, suggesting that obesity is influenced by numerous mechanisms that are not fully captured by survey behavior or state-level environmental summaries. Our comprehension of the variables with the greatest predictive power was enhanced by feature-importance analyses. In both Decision Tree and Logistic Regression models, physical activity measures were consistently among the most influential predictors. The importance of structural conditions in influencing health behaviors is highlighted by the prominence of environmental indicators such as food-access percentages, grocery and restaurant density, and fast-food sales per capita.Income and additional socioeconomic variables, along with racial indicators, mirrored the known differences in the prevalence of obesity that have been reported in public health literature. These convergent signals support a key idea: obesity develops through a combination of behavioral, social, and environmental pathways rather than being explained by individual behavior alone. The consistency of these patterns provides significant insight into population-level determinants, even though the models' moderate predictive performance restricts their application for clinical classification.

There are a few limitations that need to be noted given the project's size and design. First, measurement error is introduced by the BRFSS's reliance on self-reported data, particularly for variables pertaining to diet, physical activity, alcohol consumption, and BMI. Second, because of dataset availability, environmental variables were combined at the state level, which could result in ecological fallacies because localized food-environment patterns can differ significantly within a single state. Third, the models are unable to capture behavioral trajectories, cumulative exposures, or temporal dynamics that probably affect long-term weight status due to the cross-sectional nature of the dataset. Fourth, the discriminative power of conventional classifiers is limited by the slight class imbalance and high overlap between the predictor distributions of obese and non-obese individuals. Despite these difficulties, the project's methodological decisions were purposefully conservative: standardization, outlier exclusion, stratified cross-validation, and transparent hyperparameter tuning all sought to improve reproducibility and lessen bias within the limitations of the available data.

Beyond the scope of this study, there are still a number of promising possibilities for further study. By capturing heterogeneity in food access, walkability, and socioeconomic disadvantage, more detailed environmental data at the county, census-tract, or neighborhood level could significantly increase precision. More reliable causal inference would be possible with the modeling of temporal changes in behavior and environment made possible by longitudinal BRFSS or ACS panels. As long as interpretability is maintained for public health communicability, future research may also investigate ensemble approaches or contemporary gradient-boosted models that can capture more complex nonlinear structure. Additionally, using causal-inference frameworks like propensity-score weighting, structural equation modeling, or causal forests may shed light on which behavioral or environmental changes could significantly lower the risk of obesity. Lastly, incorporating external data sources like built-environment metrics, transportation networks, or even dietary patterns derived from social media could increase future models' effectiveness for both prediction and explanation.

Overall, this project demonstrates that pairing structural food-environment indicators with individual health behaviors provides a workable and comprehensible framework for evaluating obesity risk at scale. The modeling results confirm that significant behavioral and environmental signals can be reliably identified, even though predictive accuracy is still limited because of the complexity of the outcome and the limitations of self-reported data. These results demonstrate the potential of multi-level data integration for public health research and emphasize the need for richer, more detailed data to improve comprehension of the complex causes of obesity.

## References

Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees*. Wadsworth.

Centers for Disease Control and Prevention. (2023). *Behavioral Risk Factor Surveillance System (BRFSS)*. U.S. Department of Health and Human Services. https://www.cdc.gov/brfss/

Cover, T., & Hart, P. (1967). Nearest neighbor pattern classification. *IEEE Transactions on Information Theory, 13*(1), 21–27. https://doi.org/10.1109/TIT.1967.1053964

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.

Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 374*(2065). https://doi.org/10.1098/rsta.2015.0202

MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. In L. M. Le Cam & J. Neyman (Eds.), *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1, pp. 281–297). University of California Press.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

U.S. Census Bureau. (2023). *American Community Survey (ACS)*. https://www.census.gov/programs-surveys/acs/

U.S. Department of Agriculture, Economic Research Service. (2023). *Food Environment Atlas*. https://www.ers.usda.gov/data-products/food-environment-atlas/